BASIC ELECTRONICS (24B11EC111)

Lecture-1

VISION & MISSION

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY

Vision: To become a centre of excellence in the field of IT and related emerging areas of education, training and research comparable to the best in the world for producing professionals who shall be leaders in innovation, entrepreneurship, creativity and management.

Mission:

MISSION 1: To develop as a benchmark University in emerging technologies.

MISSION 2: To provide state-of-the-art teaching learning process and R&D environment.

MISSION 3: To harness human capital for sustainable competitive edge and social relevance.

Department Vision and Mission

(DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING)

Vision: To be a centre of excellence in education, training and research in Electronics and Communication Engineering to cultivate technically competent professionals for Industry, Academia and Society

Mission:

MISSION 1: To impart education through contemporary, futuristic and flexible curricula with innovative teaching learning methods and hands on training with well equipped Labs.

MISSION 2: To carry out cutting edge research in different areas of Electronics and Communication Engineering.

MISSION 3: To inculcate technical and entrepreneurial skills in professionals to provide socially relevant and sustainable solutions.

Program Educational Objectives-PEOs

PEO1: To provide strong foundation in Electronics and Communication Engineering to pursue professional career, entrepreneurship and higher studies.

PEO2: To develop capability to analyze, design and develop feasible **solutions to real world problems.**

PEO3: To inculcate professional ethics, managerial and communication skills to develop ingenious solutions for benefit of society and environment.

Program Outcomes (POs)

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

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5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

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9. **Individual and Team Work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning**: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSO)

These outcomes are specific to a program in addition to NBA defined POs, namely, Electronics & Communication, Computer science etc.

- **PSO1:** To identify the engineering problems and develop solutions in the area of communication, signal processing, VLSI and embedded systems.
- **PSO2**: To demonstrate proficiency in utilization of software and hardware tools along with analytical skills to arrive at appropriate solutions.

COURSE OUTCOMES (CO)

CO (Code: 24B11EC111)		COGNITIVE LEVELS
C.1	Recall the concepts of various circuit elements and Kirchhoff's laws.	Remembering Level (C1)
C.2	Understand the basics of semiconductor PN junction diodes and Op-Amp, and their applications.	Understanding Level (C2)
C.3	Apply network theorems to effectively solve complex DC circuits.	Applying Level (C3)
C.4.	Explain the operation of transistors (BJT and MOSFET) and analyze their biasing techniques.	Analyzing Level (C4)

SYLLABUS

Module No.	Title of Module	Topics in the Module	No. of Lectures
1	Basic Circuit Analysis	Kirchhoff's Laws, Voltage Divider rule, Current Divider Rule, DC circuit analysis (Nodal, Mesh), Superposition and Thevenin/Norton Theorem	10
2	PN Junction Diode and Applications	PN Junction, Biasing the PN Junction, Current–Voltage Characteristics of a PN Junction, PN Junction Diodes, Half Wave Rectifier & Full Wave Rectifier Clipper & Clamping Circuits	8
3	Zener Diode and Applications	Zener Diode and applications, Line and Load Regulations of reference circuits.	4
4	Introduction to BJT	Introduction to BJT, operation, characteristics, Biasing and Stability	6
5	Introduction to MOSFET	Introduction to MOSFET, operation, characteristics and biasing	6

SYLLABUS

Module No.	Title of Module	Topics in the Module	No. of Lectures
6	Op-amp and Applications	Block Diagram Representation of Typical Op-Amp, Schematic Symbol, Op-Amp parameters, Ideal Op-Amp, Equivalent Circuit of Op-Amp, Op-Amp Applications: Inverting Configuration, Non-Inverting Configuration, Voltage Follower, summer, comparator, difference Amplifier, Integrator, Differentiator	8
		Total no. of lectures	42

EVALUATION CRITERIA

Components	Maximum Marks
• Mid Term Examination	30
• End Term Examination	40
• TA	30 (Assignments = 7, Attendance = 10 , Quiz =5, PBL=8)
• Total	100
• Passing Marks	40
• Attendance Criteria	80

RECOMMENDED BOOKS

1. R.C. Dorf and James A. Svoboda, “Introduction to Electric Circuits”, 9th ed, John Wiley & Sons, 2013.
2. R. Boylestad and L. Nashelsky, ‘Electronic Devices and Circuit Theory’, PHI, 7e, 2001.
3. Charles K. Alexander (Author), Matthew N.O Sadiku, “ Fundamentals of Electric Circuits”, 6th ed, Tata Mc Graw Hill, 2019.
4. Sze and Ng," Physics of Semiconductor Devices", 3rd Ed., 2006.
5. Ramakant A. Gayakwad, “Op-Amps and Linear Integrated Circuits”, 4th edition, Pearson, 2000.